

# Advanced Machine Tools Lab

Subject Code 6029 lesson HTML for Mechanical Engineering. This page is a student-friendly study guide with type-specific sections, expected questions and a separate Master Answer Key.

Official syllabus reference: [SITTTR course contents](#) | Author: Nandakumar.M | Visit: [nandakumarm.dpdns.org](http://nandakumarm.dpdns.org)

# 6029 - Advanced Machine Tools Lab

Department(s): Mechanical Engineering. Semester: Semester 6. Subject type: Lab.

## 6029

**Course Code**

Advanced Machine Tools Lab

## 6

**Semester**

Semester 6

## Lab

**Study Mode**

Lab / Practical

### How to Study

- Read the overview and make a one-page summary.
- Open each module separately and practise the expected questions.
- Keep diagrams, tables, formulas or procedures neat according to subject type.
- Use the Master Answer Key only after attempting the questions.

## Study Support

Use Malayalam for classroom discussion if needed, but keep technical terms in English.  
Attempt module questions first, then check the Master Answer Key.

# Experiment Bank for Advanced Machine Tools Lab

Quick reference cards for this subject type.

## Safety Rules

Switch off supply before wiring, use correct range, avoid loose connections and follow lab instructions.

## Apparatus List

Prepare instruments, trainer kits, tools, meters and consumables required for Advanced Machine Tools Lab.

## Record Format

Aim, apparatus, theory, diagram, procedure, observation, calculation, result and precautions.

## Viva Focus

Principle, instrument use, range, error sources, precautions and applications.

## Practical Exam Tips

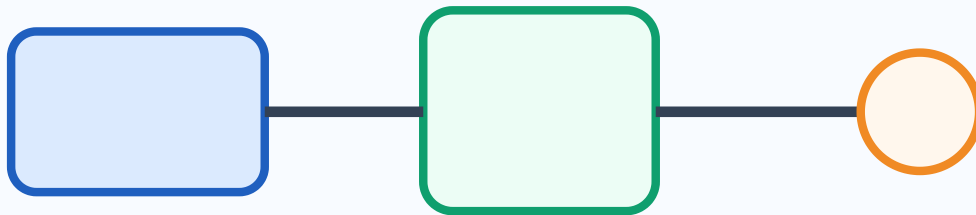
Neat circuit/setup diagram, correct observation table and clear final result.

# Lab Safety and Instruments

Identify tools, instruments, ratings, symbols and safety precautions.

## Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Mechanical Engineering applications.



**Aim -> Apparatus -> Procedure -> Observation -> Result**

Visual study block for Advanced Machine Tools Lab

## Simple Explanation

Identify tools, instruments, ratings, symbols and safety precautions. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

## Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

## Expected Questions

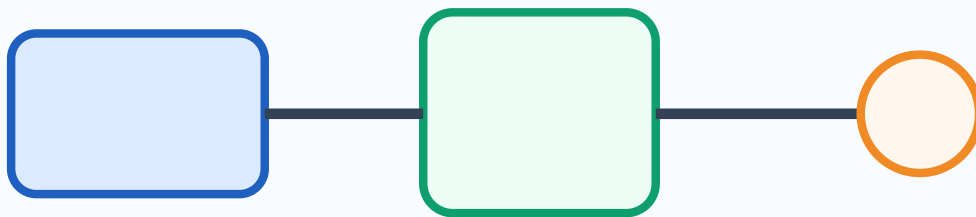
- 1 List the safety precautions for this lab.
- 2 Write the apparatus list format for an experiment.
- 3 Explain how to choose instrument range.
- 4 Write the record format for practical work.

# Basic Experiments and Setup

Prepare aim, setup/circuit diagram, procedure and observations.

## Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Mechanical Engineering applications.



**Aim -> Apparatus -> Procedure -> Observation -> Result**

Visual study block for Advanced Machine Tools Lab

## Simple Explanation

Prepare aim, setup/circuit diagram, procedure and observations. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

## Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

## Expected Questions

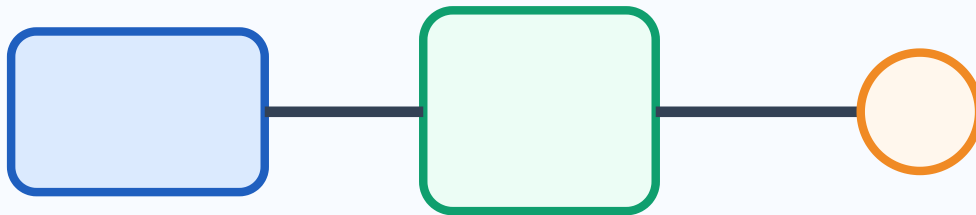
- 1 Write the aim and procedure format for a basic experiment.
- 2 Draw a neat setup/circuit block for a practical.
- 3 Prepare an observation table model.
- 4 List common connection/setup mistakes.

# Observation, Calculation and Result

Record readings, calculate values and write final result correctly.

## Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Mechanical Engineering applications.



**Aim -> Apparatus -> Procedure -> Observation -> Result**

Visual study block for Advanced Machine Tools Lab

## Simple Explanation

Record readings, calculate values and write final result correctly. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

## Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

## Expected Questions

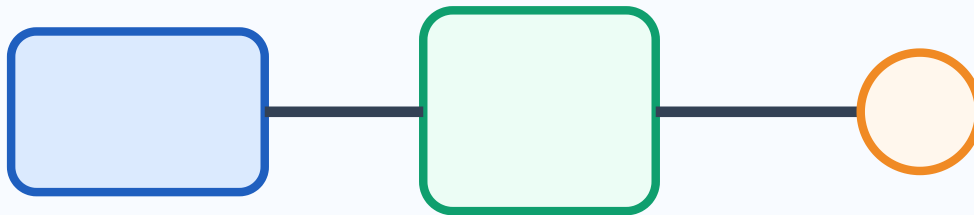
- 1 Explain how to fill an observation table.
- 2 Write a calculation format with units.
- 3 State how to write result and inference.
- 4 List error sources and precautions.

# Viva and Practical Examination

Revise viva questions, troubleshooting and exam record presentation.

## Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Mechanical Engineering applications.



**Aim -> Apparatus -> Procedure -> Observation -> Result**

Visual study block for Advanced Machine Tools Lab

## Simple Explanation

Revise viva questions, troubleshooting and exam record presentation. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

## Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

## Expected Questions

- 1 List five viva questions from this lab.
- 2 Write precautions for practical examination.
- 3 Explain how to troubleshoot a wrong reading.
- 4 Write record submission checklist.

# One-page Quick Revision

Last-minute checklist for Advanced Machine Tools Lab.

## Important Points

- Definitions and key terms.
- Diagrams, formulas, procedures or formats.
- Applications in Mechanical Engineering.

## Exam Method

- OK** Read question carefully.
- OK** Write answer in steps.
- OK** Label diagrams and units.
- OK** Finish with result/application.

## Common Mistakes

- NO** Skipping units or labels.
- NO** Writing answer without structure.
- NO** Mixing definitions and applications.
- NO** Leaving diagrams untidy.

# Answers and Hints

Answer hints exactly match the expected questions inside modules.

## Module 1 Answer Hints

### M1-Q1

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M1-Q2

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M1-Q3

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M1-Q4

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

## Module 2 Answer Hints

### M2-Q1

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M2-Q2

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M2-Q3

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M2-Q4

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

## Module 3 Answer Hints

### M3-Q1

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M3-Q2

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M3-Q3

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M3-Q4

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

## Module 4 Answer Hints

### M4-Q1

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M4-Q2

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M4-Q3

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.

### M4-Q4

Write the answer in practical-record style: aim, apparatus, principle, procedure, observation/calculation, result and precautions. Connect the point directly with Advanced Machine Tools Lab.